

Exercise 2.6. Determine conditions on integers m and n for which mn is even. Write down your conditions as a conjecture, and then prove that your conjecture is correct.

Solution.

The question asks us to find the conditions on m and n that make the product mn even, state that as a conjecture, and then prove it.

Step 1: Look at the possibilities

Every integer is either even or odd.

There are four possible combinations:

m	n	mn
Even	Even	Even
Even	Odd	Even
Odd	Even	Even
Odd	Odd	Odd

So we can make the following conjecture.

Conjecture

mn is even if and only if at least one of m or n is even.

In other words,

mn is even $\iff m$ is even or n is even.

Proof

We need to prove **both directions**.

Direction 1: If m or n is even, then mn is even.

Suppose m is even.

Then, by definition,

$$m = 2k$$

for some $k \in \mathbb{Z}$.

Therefore,

$$\begin{aligned} mn &= (2k)n \\ &= 2(kn). \end{aligned}$$

Since kn is an integer, mn has the form $2r$.

Therefore,

mn is even.

The same argument works if n is even.

Direction 2: If mn is even, then m or n is even.

We prove this using the contrapositive.

The original statement is:

$$mn \text{ is even} \Rightarrow m \text{ or } n \text{ is even.}$$

Its contrapositive is:

$$m \text{ and } n \text{ are odd} \Rightarrow mn \text{ is odd.}$$

Suppose m and n are both odd.

Then

$$m = 2a + 1$$

and

$$n = 2b + 1$$

for some $a, b \in \mathbb{Z}$.

Then

$$mn = (2a + 1)(2b + 1).$$

Expanding:

$$mn = 4ab + 2a + 2b + 1.$$

Factor out 2:

$$mn = 2(2ab + a + b) + 1.$$

This has the form

$$2k + 1,$$

so mn is odd.

Thus, if mn is even, m and n cannot both be odd. Therefore, at least one of them must be even.

$$\boxed{mn \text{ is even} \Rightarrow m \text{ or } n \text{ is even.}}$$

Combining both directions:

$$\boxed{\therefore mn \text{ is even if and only if at least one of } m, n \text{ is even.}}$$